Flexible RISs: Learning-based Array Manifold Estimation and Phase-shift Optimization

Mohamadreza Delbari, Ehsan Mohammadi, Mostafa Darabi, Arash Asadi, Alejandro Jiménez-Sáez, and Vahid Jamali

TU Darmstadt
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Mohamadreza Delbari
Technical University of Darmstadt

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Outline



1- Introduction

2- Modeling

3- Problem formulation and Solution

4- Performance Evaluation

5- Conclusions

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Introduction
Motivation

- Demands:
  Higher data rates and capacity are required in modern communication systems (e.g., 5G, 6G).
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Time	XG	Aim	Data rate
1979	1G	Voice	2 - 3 kbps
1991	2G	SMS	20 - 40 kbps
1998	3G	Data	0.5 - 3 Mbps
2009	4G	Streaming	20 – 100 Mbps
2019	5G	Sensing	100 – 400 Mbps
2030	6G	Intelligence	1 – 10 Gbps

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Intelligence $\rightarrow$ RIS

Introduction

RIS Implementation and Wavelength Scaling

- **Frequency Scaling:** Higher carrier frequency leads to sub-centimeter wavelength (λ).
- **Sensitivity:** Minor structural displacements or surface curvature ($\approx \lambda$) cause severe phase mismatch.
- **Goal:** Enable flexible RIS deployment on arbitrary structures (e.g., pillars, vehicles, curved facades) [1-3].

Band	Frequency	Wavelength (λ)	Spacing ($\lambda/2$)
Sub-6 GHz	3.5 GHz	85.7 mm	42.8 mm
mmWave	28 GHz	10.7 mm	5.3 mm
mmWave	60 GHz	5.0 mm	2.5 mm
Sub-THz	140 GHz	2.1 mm	1.0 mm

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Do not confuse position optimization with flexible RIS!
[4-5] $\rightarrow$ Movable antennas $\rightarrow$ Tutorial Prof. Björnson

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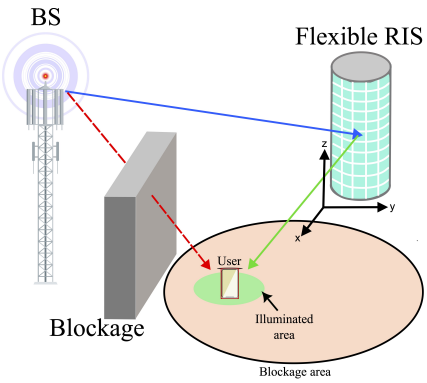
3- Problem formulation and Solution

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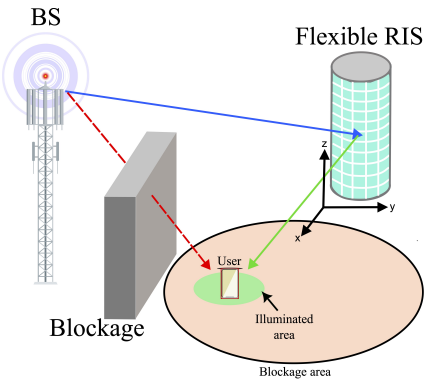
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Modeling
System Model



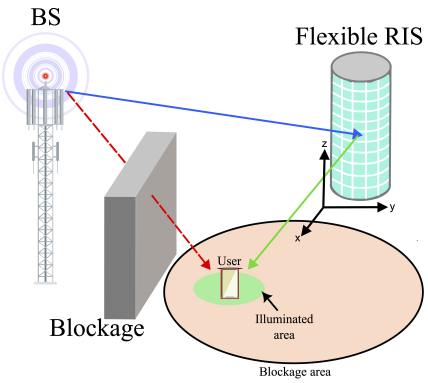
Modeling
System Model



$$y_k = \left(\mathbf{h}_{d,k}^{\mathsf{H}} + \mathbf{h}_{r,k}^{\mathsf{H}} \mathbf{\Gamma} \mathbf{H}_t\right)\mathbf{x} + n_k,$$

y_k :	Received signal,
$\mathbf{x}$:	Transmitted signal,
n_k :	Noise at MU k,
$\mathbf{h}_{d,k}^{\mathsf{H}}$:	BS-MU k channel,
$\mathbf{h}_{r,k}^{\mathsf{H}}$:	RIS-MU k channel,
$\mathbf{H}_t^{\mathsf{H}}$:	BS-RIS channel,
$\mathbf{\Gamma}$:	RIS Phase shifts,

Modeling
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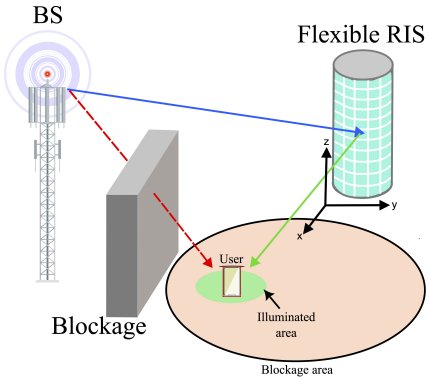


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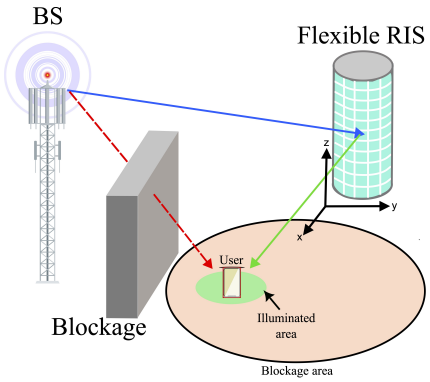
$$\mathbf{H} = \sqrt{\frac{K_f}{K_f+1}}\mathbf{H}^{\text{LOS}} + \sqrt{\frac{1}{K_f+1}}\mathbf{H}^{\text{nLOS}},$$

Modeling
RIS Model



$$x = g(y,z) \approx \sum_{m_y=0}^{M_{\text{total}}} \sum_{m_z=0}^{M_{\text{total}}-m_y} a_{m_y,m_z} y^{m_y} z^{m_z},$$

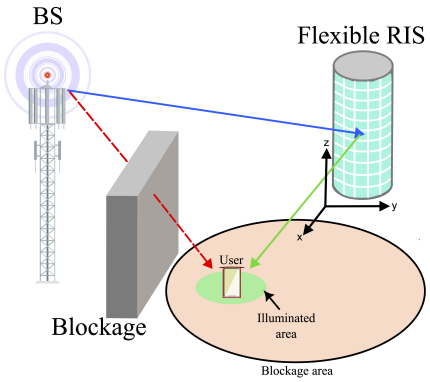
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$$\frac{1}{2}\frac{(M_{\text{total}}+2)!}{M_{\text{total}}!} \ll N$$

M_{total}	2	3	4
Parameters	6	10	15

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Problem formulation and Solution

Problem formulation

The noiseless received power via the LOS path is

$$P(k,\boldsymbol{\omega};g)=\left|\alpha_0\int_{-\frac{L_y}{2}}^{\frac{L_y}{2}}\int_{-\frac{L_z}{2}}^{\frac{L_z}{2}}\mathrm{e}^{\mathrm{j}\omega(y,z)+\mathrm{j}\kappa d_k(y,z;g)}\mathrm{d}y\mathrm{d}z\right|^2,$$

$$\hat{g}=\arg\min_{g}\quad\sum_{k=1}^{K}\sum_{m=1}^{M}\left\|P_{k,m}-P(k,\boldsymbol{\omega};g)\right\|^2.$$

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k \ $\boldsymbol{\omega}$	$\boldsymbol{\omega}_1$	$\boldsymbol{\omega}_2$	$\dots$	$\boldsymbol{\omega}_M$
1	$P_{1,1}$	$P_{1,2}$	$\dots$	$P_{1,M}$
2	$P_{2,1}$	$P_{2,2}$	$\dots$	$P_{2,M}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
K	$P_{K,1}$	$P_{K,2}$	$\dots$	$P_{K,M}$

Table: Received power measurements ($P_{k,m}$) across K user locations and M different RIS phase configurations.

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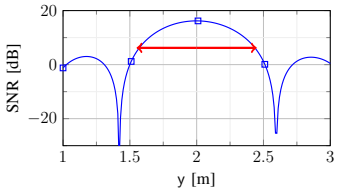
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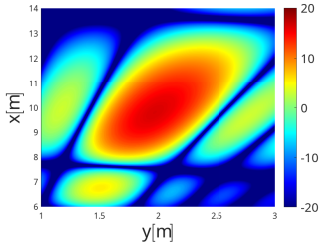
Solutions → Conformal Manifolds → Non-Linear Least Squares → Deep Learning

Problem formulation and Solution

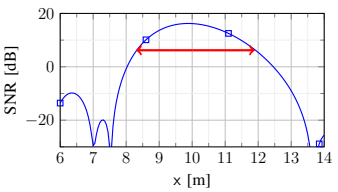
Independent Measurements



(a) The received power along to the y axis.



(b) The received power inside the coverage area.



(c) The received power along to the x axis.

Figure: The required minimum distance for the spatial density of the measurements ($N_y = 40, N_z = 10$).

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Performance Evaluation

Simulation Setup

Parameter	Value
Frequency	28 GHz
BS Antennas (x − z axes)	4×4
RIS Elements (y − z-axes)	40×10
Channel Model	Ricean
Path Loss Model	$(c/(4\pi f_c d))^2$
Independent samples	18225
Training Set Size	80%
Validation Set Size	10%
Test Set Size	10%

Network Architecture	
Hidden Layers	3 (Dense)
Neuron Configuration	$128 \rightarrow 64 \rightarrow 32$
Hidden Activation	ReLU
Output Activation	Linear
Training Configuration	
Optimizer	Adam
Learning Rate	0.001
Loss Function	MSE
Maximum Epochs	500
Batch Size	100
Early Stopping Patience	50 Epochs

Performance Evaluation

Convergence

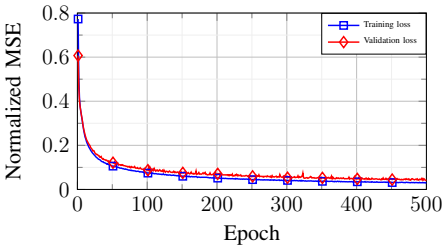


Figure: Training loss and validation loss versus the number of epochs.

Performance Evaluation

Geometry Estimation

$$x = g(y,z) \approx a_{yy}y^2 + a_{zz}z^2 + a_{yz}yz + a_y y + a_z z.$$

$a_{yy}, a_{zz}, a_{yz} \in \mathcal{U}(0, \sigma)$
$a_y, a_z \in \mathcal{U}(-\frac{\sigma}{10}, \frac{\sigma}{10})$

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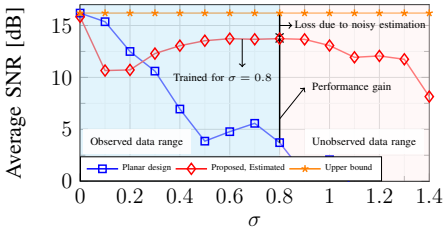
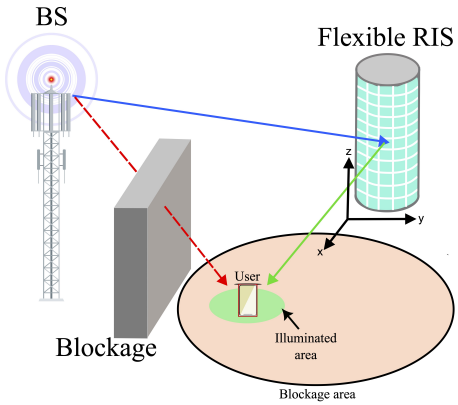


Figure: Rx power at the MU versus the geometric parameter variance.

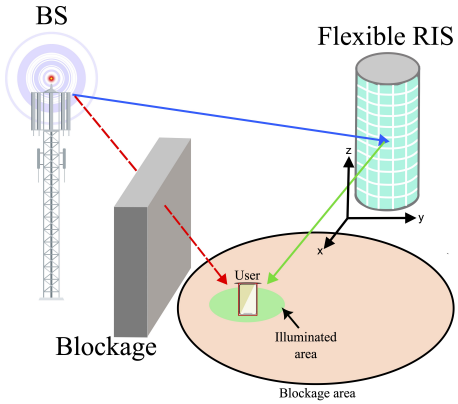
Performance Evaluation

Localization Error



Performance Evaluation

Localization Error



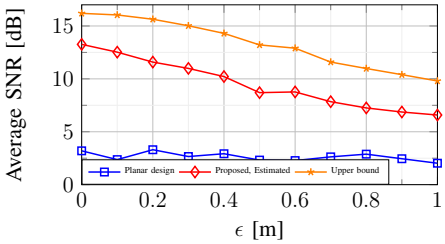


Figure: The Rx power in MU versus the variance of the localization error, $\mathcal{U}(-\epsilon, \epsilon)$.

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Summary and Takeaways

Summary:

- Flexible RISs have the benefit of conforming to arbitrary surfaces.
- When the RIS is non-planar, phase-shift optimization becomes challenging due to unknown geometry.
- We proposed a solution based on power measurements using deep learning to estimate surface geometry.
- The trained deep learning framework remains robust even in the presence of user localization errors.

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Future Research Directions:

- Design DL architectures operating robustly without accessing to all measurement points.

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Thank you for your attention!

Questions?

Email: mohamadreza.delbari@tu-darmstadt.de



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